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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING – (24CI3201)
A.Y. 2026 - 2027

TITLE OF THE PROJECT		
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PROJECT TITLE
Adaptive AI Research Intelligence Platform for Automated Literature Review and Research Gap Discovery

ABSTRACT

The Adaptive AI Research Intelligence Platform is an intelligent research assistance system designed to automate and simplify the process of literature review and research gap discovery. The system collects and analyzes relevant research papers from academic sources based on user-provided research topics, keywords, or queries. It uses Artificial Intelligence, Natural Language Processing, and Machine Learning techniques to extract important information from research papers, including abstracts, methodologies, datasets, findings, limitations, and citations. Unlike traditional literature review approaches that require researchers to manually search, read, compare, and organize a large number of research papers, the proposed system automatically processes and summarizes relevant literature to provide a structured understanding of the existing research landscape.

The platform uses semantic search and Natural Language Processing techniques to identify relationships and similarities between research papers, categorize papers according to research themes, and rank them based on their relevance to the user's research topic. Large Language Models are utilized for intelligent summarization, question answering, and contextual analysis of research literature. The system further analyzes the findings, methodologies, limitations, and future-scope sections of selected papers to identify unexplored areas, inconsistencies, limitations, and potential research opportunities. These insights are used to generate potential research gaps and suggest possible directions for future research.

The system is developed using React.js for the interactive frontend, Node.js for backend services and API management, Python for Natural Language Processing, Machine Learning, semantic analysis, and research-gap detection, and MongoDB for storing user profiles, research papers, search history, and analysis results. A vector database such as FAISS or ChromaDB is used for storing and retrieving semantic embeddings of research documents. Academic research APIs are integrated to retrieve relevant research publications, while PDF processing libraries are used to extract and analyze text from research papers. The recommendation and analysis components can continuously improve based on user searches, selected papers, feedback, and research interests, enabling the platform to adapt to individual research requirements over time.

The proposed platform aims to reduce the time and effort required for systematic literature exploration, improve the relevance and organization of research information, and assist researchers in discovering meaningful research gaps. It provides a scalable architecture in which the frontend, backend, database, AI models, document-processing pipeline, and semantic search components work together efficiently. The project demonstrates how Artificial Intelligence and modern web technologies can be integrated to develop an adaptive research intelligence platform that supports researchers in literature analysis, knowledge discovery, and identification of potential future research directions.

Tools and Technologies Used

Frontend: React.js / Next.js, HTML, CSS, JavaScript

Backend: Node.js, Express.js

AI/ML: Python, PyTorch, Scikit-learn

Natural Language Processing: Hugging Face Transformers, Sentence Transformers, spaCy, NLTK

Large Language Model: Gemini API / OpenAI API

Semantic Search: Sentence Transformers, FAISS / ChromaDB

Database: Supabase

Academic Research APIs: Semantic Scholar API, OpenAlex API, Crossref API

PDF Processing: PyMuPDF / pdfplumber

Data Visualization: Recharts / Plotly

Authentication: JWT / OAuth

Version Control: Git and GitHub

Deployment: Vercel, Render / Railway

Signature of the Guide HOD